Pharmaceutical Sales Trend Analysis

Power BI portfolio project · Daily pharmacy sales data, 2014–2019 · 8 drug categories

1. Executive Summary

This report analyses 127.57K packages sold across eight drug categories between 2014 and 2019. The Power BI dashboard covers category mix, monthly and weekday seasonality, year-over-year (YoY) movement, volatility, cross-category correlation and anomaly detection.

Key findings

- **Anilides dominate.** They account for about 49% of all packages sold (63K), more than three times the next category.
- **Sales are strongly seasonal.** Anilides peak in January (7.6K) and fall by roughly half to a mid-year trough in June–July (3.6–3.7K), with a second peak in October (6.8K).
- **Weekends lead.** Saturday is the strongest day (19.8K) and Thursday the weakest (17.2K), but the spread is only about 15%.
- **Demand is volatile year to year.** Totals rose through 2016 (25.2K), fell 23% in 2017, and recovered in 2018 (22.9K).
- **Categories move largely independently.** All pairwise correlations are weak (maximum 0.29).
- **Anomalies are rare.** 412 flagged records make up 2.45% of all records, and they are mainly unusually low-sales days.

2. Headline KPIs

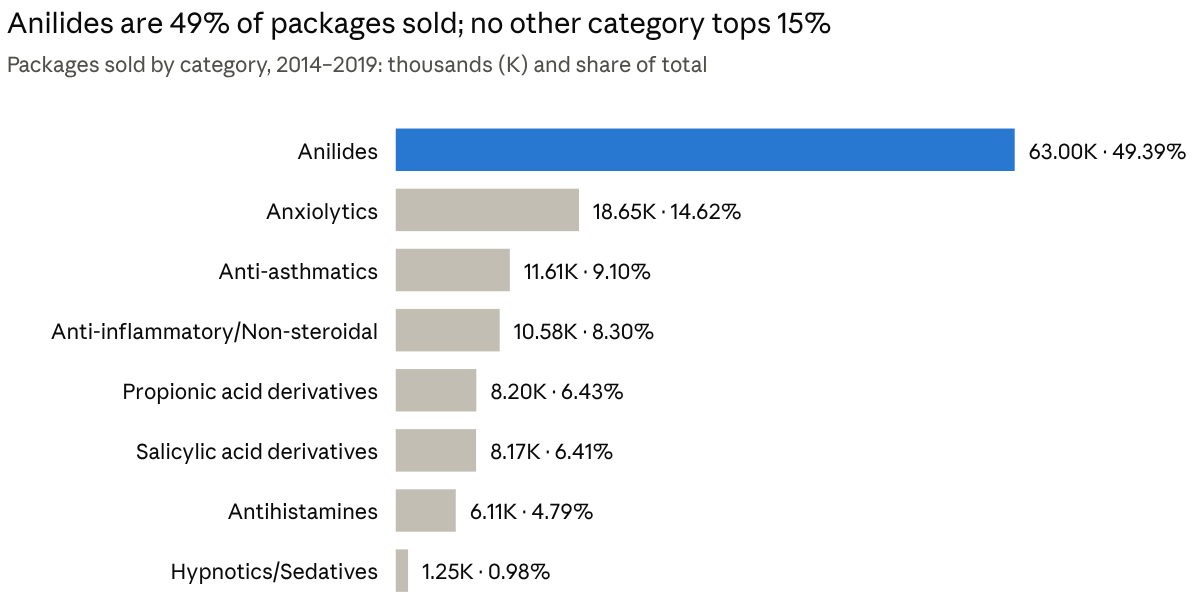
Metric	Value	Interpretation
Total packages sold	127.57K	Sum across all categories, 2014–2019
Average daily sales	7.57	Average per category per day
Sales volatility	10.90	Standard deviation of daily sales; larger than the mean, so sales are highly dispersed
Anomaly count	412	Records flagged as anomalous

Metric	Value	Interpretation
Anomaly share	2.45%	0.41K anomalous vs 16.44K normal records

The volatility figure (10.90) exceeds the average (7.57), a sign of a right-skewed distribution. A few high-volume categories, chiefly Anilides, drive most of the spread.

3. Category Performance

Rank	Category	Packages	Share
1	Anilides	63.00K	49.39%
2	Anxiolytics	18.65K	14.62%
3	Anti-asthmatics	11.61K	9.10%
4	Anti-inflammatory/Non-steroidal	10.58K	8.30%
5	Propionic acid derivatives	8.20K	6.43%
6	Salicylic acid derivatives	8.17K	6.41%
7	Antihistamines	6.11K	4.79%
8	Hypnotics/Sedatives	1.25K	0.98%



Power BI dashboard: sales distribution of categories · all 8 categories, 2014–2019

Observations

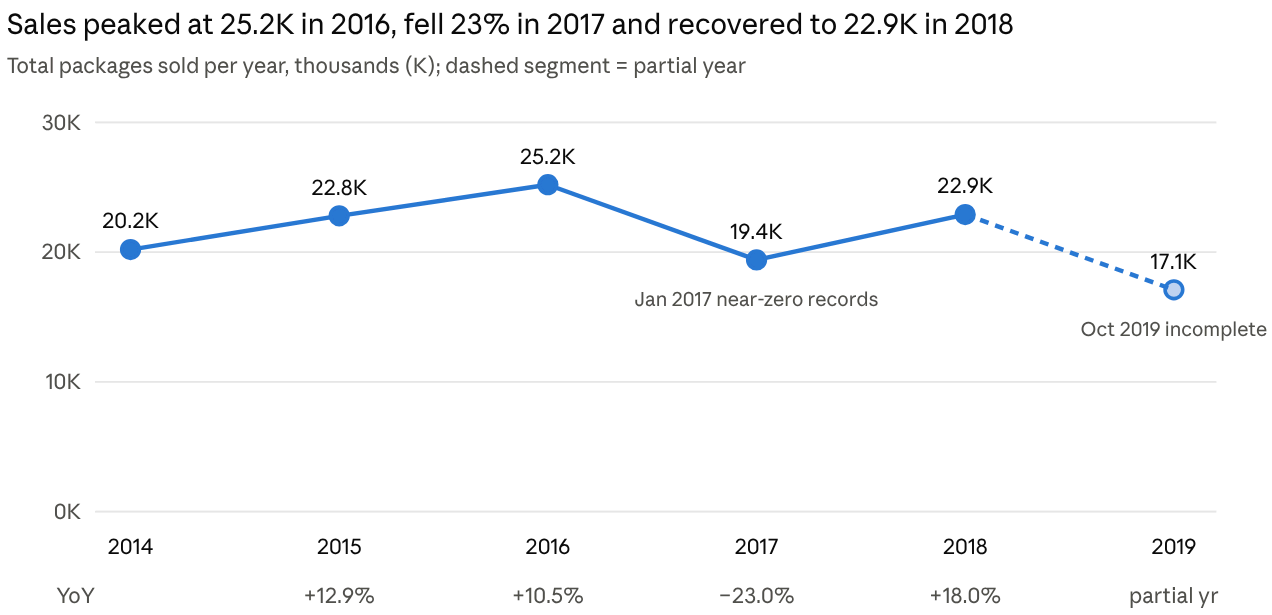
- The top two categories (Anilides and Anxiolytics) together make up about 64% of volume.
- The middle four categories (ranks 3–6) are fairly close, between 6% and 9% each.
- Hypnotics/Sedatives are a niche line at under 1% of volume.
- Concentration is a business risk. Any supply, pricing or regulatory change affecting Anilides would hit total volume far more than any other category.

4. Time-Series Analysis

4.1 Yearly trend and YoY change

Year	Total packages	YoY change
2014	20.2K	n/a
2015	22.8K	+12.9%
2016	25.2K	+10.5%
2017	19.4K	−23.0%

Year	Total packages	YoY change
2018	22.9K	+18.0%
2019	17.1K	Not comparable (partial year)



Power BI dashboard: monthly sales trend by year · 2019 is a partial year

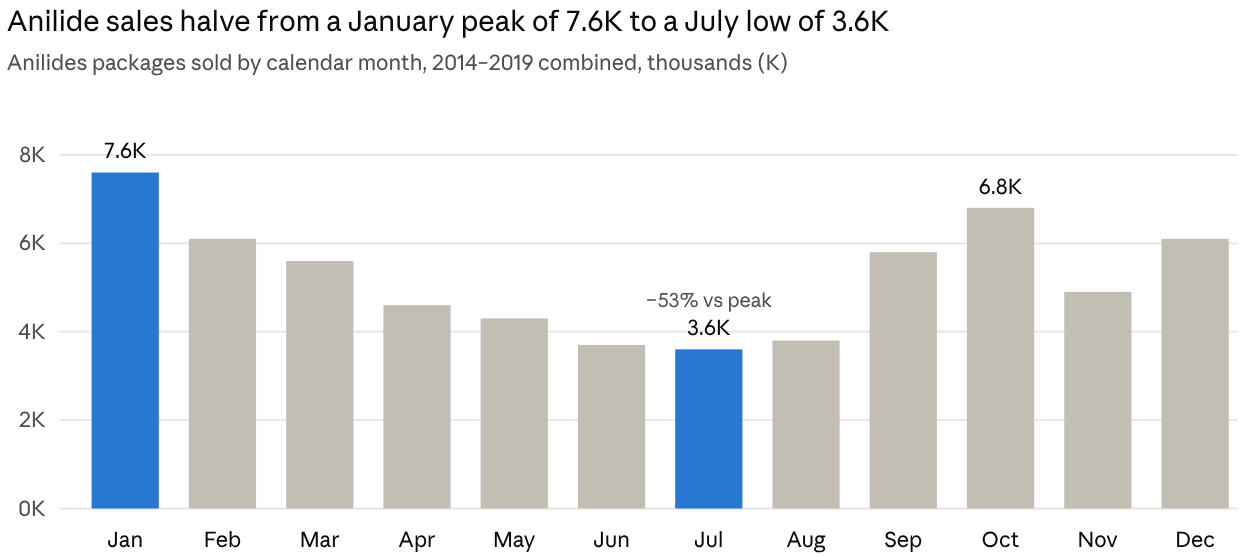
- Sales grew for two consecutive years and peaked in 2016.
- The 2017 drop is partly a data-quality effect, since January 2017 contains near-zero records.
- 2019 is an incomplete year (October is only partially captured), so its lower total should not be read as a decline.

Anilides by year: 9.3K (2014), 11.9K (2015), 13.3K (2016), 9.3K (2017), 11.2K (2018). They track the overall trend almost exactly, which confirms they are the main driver of total movement. Anxiolytics started at roughly 4.3K in 2014 and settled around 2.6–3.3K in later years, so they are one of the few categories with a clear structural decline.

4.2 Monthly seasonality

Anilides, summed across all years:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Packages (K)	7.6	6.1	5.6	4.6	4.3	3.7	3.6	3.8	5.8	6.8	4.9	6.1



Power BI dashboard: Anilides by month · values summed across 2014–2019

- **Winter peak:** January is the highest month, then volumes decline steadily through spring.
- **Summer trough:** June–July is the low point, roughly 53% below January.
- **Autumn rebound:** a sharp recovery from September, peaking in October, followed by a dip in November and a December bounce.
- **Other categories** show much flatter profiles. Anxiolytics stay in a band of roughly 1.2–2.0K per month, and most remaining categories sit below about 1.3K. Hypnotics/Sedatives are essentially flat at around 0.1K.

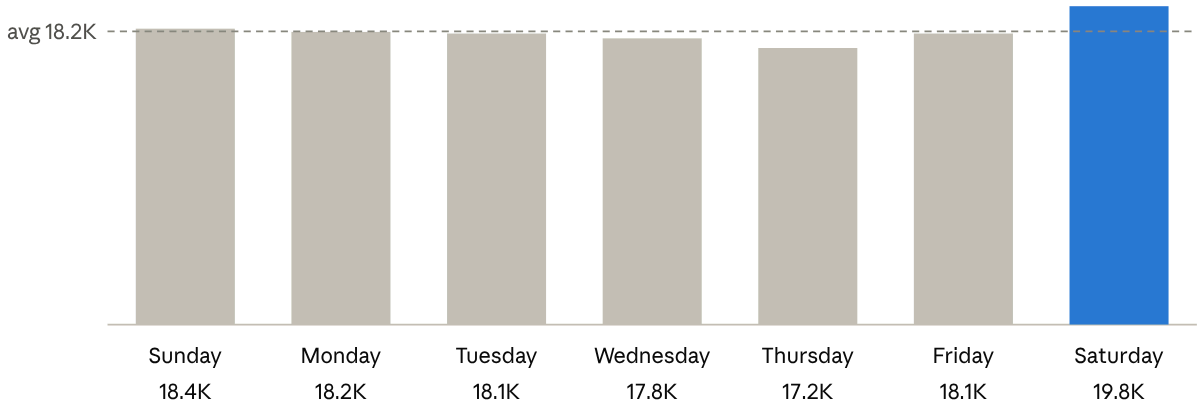
The Anilides pattern is consistent with seasonal conditions such as winter illness and the post-monsoon/autumn period. This is a hypothesis to test, not something the data proves.

4.3 Day-of-week pattern

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Packages (K)	18.4	18.2	18.1	17.8	17.2	18.1	19.8

Saturday is the busiest day at 19.8K, about 15% above Thursday's 17.2K

Packages sold by day of the week, 2014–2019 combined, thousands (K); bars start at zero



Power BI dashboard: sum of packages sold by day of the week · 2014–2019

- Saturday is the busiest day (about 15.5% of volume), and Thursday is the quietest (about 13.5%).
- Anilides show a pronounced Sunday spike (10,051 packages vs about 8,400–8,800 on Monday to Friday).
- Anxiolytics are steadier through the week, ranging from about 2.6K to 3.0K on weekdays.
- The weekly cycle is mild. Day-of-week matters far less than month-of-year for planning.

5. Volatility and Risk Profile

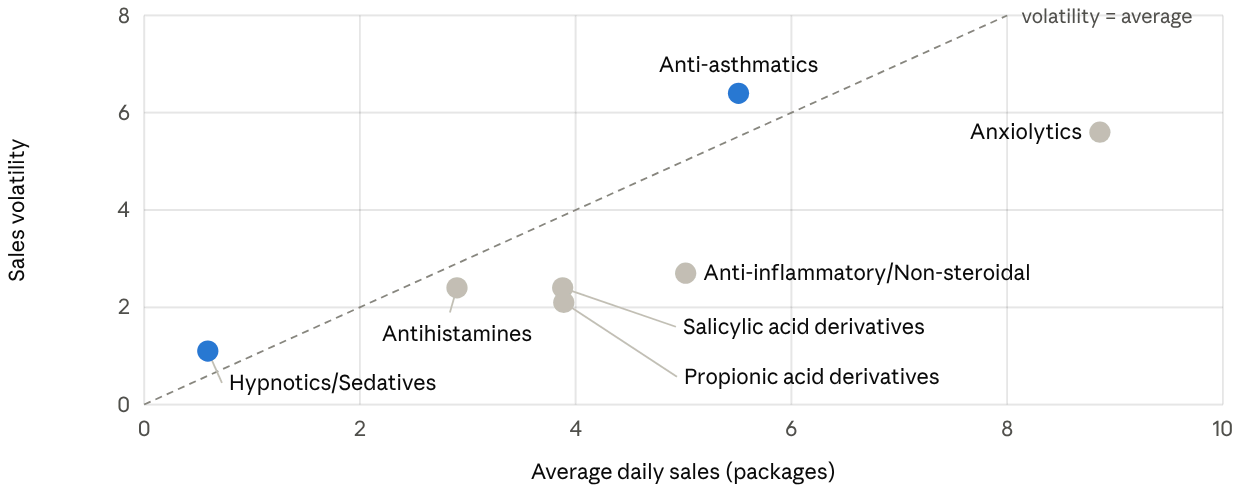
The scatter plot of average daily sales against sales volatility groups the categories as follows:

Profile	Categories	Reading
High volume, high volatility	Anilides (implied ~30/day from totals; off the visible axis range), Anxiolytics (~8.8/day, volatility ~5.6)	Largest revenue drivers; hardest to forecast precisely
Mid volume, high volatility	Anti-asthmatics (~5.5/day, volatility ~6.3)	Volatility exceeds average sales, so demand is spiky

Profile	Categories	Reading
Mid and low volume, stable	Anti-inflammatory, Propionic acid and Salicylic acid derivatives, Antihistamines (~3–5/day, volatility ~2–2.7)	Predictable, steady sellers
Very low volume	Hypnotics/Sedatives (~0.6/day, volatility ~1.2)	Small niche, but volatile relative to its size

Volatility exceeds average daily sales for Anti-asthmatics and Hypnotics/Sedatives

Average daily sales against sales volatility, by category (per category-day)



Anilides (about 30 a day) lie beyond the dashboard's axis and are not plotted.
Volatility values are read off the dashboard chart, so approximate.

Power BI dashboard: average daily sales and volatility by attribute · volatility read from the chart (approximate)

Anti-asthmatics are the notable outlier, with volatility higher than their own average. Their demand is likely event- or season-driven, which makes safety stock harder to set.

6. Cross-Category Correlation

The correlation matrix shows weak relationships throughout (all coefficients between −0.13 and 0.29 outside the diagonal).

Pair	Correlation	Note
Anilides – Propionic acid derivatives	0.29	Strongest pair in the matrix

Pair	Correlation	Note
Anxiolytics – Hypnotics/Sedatives	0.25	Plausible link: both used for anxiety and sleep-related conditions
Anilides – Anti-asthmatics	0.22	Weak positive
Anilides – Salicylic acid derivatives	0.21	Weak positive
Propionic acid – Anti-inflammatory	0.19	Related analgesic/anti-inflammatory classes
Anilides – Antihistamines	−0.13	Only meaningful negative relationship

Categories move largely independently: no pair correlates above 0.29

Correlation between category sales, 2014–2019

	1	2	3	4	5	6	7	8
1. Anilides	1.00	0.22	-0.13	0.18	0.07	0.00	0.29	0.21
2. Anti-asthmatics	0.22	1.00	0.01	0.08	0.04	0.01	0.09	0.04
3. Antihistamines	-0.13	0.01	1.00	0.09	-0.06	0.00	0.03	-0.03
4. Anti-inflammatory/Non-steroidal	0.18	0.08	0.09	1.00	0.03	0.01	0.19	0.15
5. Anxiolytics	0.07	0.04	-0.06	0.03	1.00	0.25	0.06	0.18
6. Hypnotics/Sedatives	0.00	0.01	0.00	0.01	0.25	1.00	0.02	0.06
7. Propionic acid derivatives	0.29	0.09	0.03	0.19	0.06	0.02	1.00	0.17
8. Salicylic acid derivatives	0.21	0.04	-0.03	0.15	0.18	0.06	0.17	1.00

Columns follow the row order (1–8). Blue = positive, red = negative; shading is scaled so 0.3 is darkest.

Power BI dashboard: category correlation matrix · all 8 categories

Takeaways

- No category is a reliable proxy for another, so each needs its own forecast rather than a shared driver.
- The weak positive links among analgesic and anti-inflammatory groups hint at modest shared demand drivers, such as pain and fever seasons.
- Antihistamines move slightly against Anilides, which may reflect a different seasonal cycle (allergy rather than infection-driven demand).
- Correlation at this level does not imply causation or substitution.

7. Anomaly Detection

- **412 records (2.45%)** are flagged as anomalies, against 16.44K normal records (97.55%).
- On the daily scatter for early 2014, flagged points cluster at low values (about 5–25 packages), well below the normal band (about 30–105).
- Anomalies therefore appear to be **demand drops or recording gaps** rather than spikes.
- At least one normal-flagged point sits at zero in early January 2014, which suggests the rule may miss some extreme lows. The rule should be reviewed.

Suggested follow-up: reconcile flagged dates against stock-outs, holidays, public-health events and data-entry gaps before treating them as true demand signals.

8. Data Quality and Limitations

- **Partial 2019:** the final year is incomplete, so annual comparisons with 2019 are unreliable.
- **Near-zero January 2017:** these records distort the 2017 total and the 2017 YoY figure.
- **Mixed date formats:** dates arrived in mixed formats and were standardised in Power Query.
- **Short history:** five full years limit the reliability of seasonality estimates and long-range forecasting.
- **No price or revenue data:** the analysis is in packages, not currency, so category value ranking may differ from volume ranking.
- **No external drivers:** weather, promotions, supply constraints and disease incidence are not in the dataset.
- **Source:** public pharmaceutical sales dataset from Kaggle (Milan Zdravkovic), aggregated at daily, weekly and monthly levels.

9. Recommendations

1. **Plan inventory around seasonality.** Build stock for Anilides ahead of September–October and January, and run lean in June–July.
2. **Diversify the category mix.** Review whether growth initiatives for Anxiolytics, Anti-asthmatics and the analgesic groups can reduce reliance on a single category at about 49% of volume.

3. **Use per-category forecasts.** Weak correlations argue against one shared model. Fit separate seasonal models, using a seasonal approach such as Seasonal Naïve, ETS or SARIMA, for the top three categories.
4. **Set safety stock by volatility.** Anti-asthmatics and Anxiolytics need larger buffers relative to their averages.
5. **Staff and replenish for weekends.** Align deliveries and staffing with the Saturday peak and Sunday Anilides spike.
6. **Fix data quality first.** Correct or exclude the January 2017 near-zeros and confirm October 2019 completeness before finalising any forecast.
7. **Investigate the Anxiolytics decline.** Determine whether the drop since 2014 reflects market change, substitution or data issues.

10. Next Steps

- Add a forecasting page (12-month outlook with confidence bands) in Power BI.
- Add a YoY and month-on-month variance view with small multiples by category.
- Refine the anomaly rule (rolling median or IQR per category) and document thresholds.
- Layer in price data, if available, to convert volume insights into revenue impact.

Appendix: Dashboard Visuals Covered

1. Sum of packages sold by month and attribute (line)
2. Sum of packages sold by weekday and attribute (area)
3. Category trends over years (area)
4. Sales distribution of categories (donut)
5. Average daily sales vs sales volatility by attribute (scatter)
6. Category correlation matrix (heatmap)
7. Daily sales by anomaly flag (scatter)
8. KPI summary page: totals, monthly trend, category bars, anomaly share, weekday bars